

Individual Reflection

I. Introduction

The objective of the module was to develop an understanding of fundamental ontological design principles, with a focus on their application to AI-driven software development, and to intelligent automated software agents in particular. To that end, the module revolved around two main assignments: a case study review and an ontology design modelling exercise.

The module overall, along with both assignments, are addressed here within the framework of the Gibbs' Reflective Cycle ("the cycle"), a standard heuristic device for personal reflection on the learning process (Praveena et al, 2025). Summarized, the cycle follows six steps: Description, Feelings, Evaluation, Analysis, Conclusion, and Action Plan (Praveena et al, 2025).

Accordingly, each step of the cycle is discussed in order:

II. Description

The Description step of the Gibbs' Reflective Cycle summarizes the course of learning undertaken (Praveena et al, 2025). For this module's assignments:

1. The case study review focused on evaluating different approaches for creating an ontology for a subject matter domain; in particular, an ontology to support text mining and text-based searches across agricultural reference works (Bullen, 2025; Malik, Sharan and Hijam, 2015).
2. The ontology modelling exercise focused on developing a sample ontology for a library collection, to be used as the conceptual framework for an AI-driven search engine for library assets (Bullen, 2026). The ontological model was

composed using the Protégé 5.6.7 software platform (Bullen, 2026; Stanford Center for Biomedical Informatics, 2019).

III. Feelings

The Feelings step of the Gibbs' Reflective Cycle examines subjective motivation to engage with the subject matter of the work undertaken (Praveena et al, 2025). Overall, I felt that the module was a thorough, well-composed, and productive introduction to ontological principles and ontology model development. I found learning more about the overlap between ontological design structures and their practical reification in code as database schemas and as the foundation for inter-platform communication protocols, in particular, to be particularly interesting.

IV. Evaluation

The Evaluation step reviews which parts of the work garnered successful outcomes, versus those that presented learning challenges or other issues (Praveena et al, 2025).

In terms of learning outcomes, I thought that the module was successful in presenting a broad overview of how to approach ontological questions, and successful in correlating the more abstract principles of ontological model development to real world software development issues. After over a decade of working with database schemas in my career role as a back end developer, in particular, it was helpful to focus on the principles and formal logic that underpin such schemas. Doing so brought into focus the fact that real world database implementations are, in broad terms, simply abstract ontological models that have been adapted or reified for practical use for a specific purpose in a given subject matter domain.

In terms of learning challenges, since I had not use the Protégé software platform before, there was a noticeable learning curve to develop enough proficiency with the mechanics of using Protégé to be able to complete the second modelling assignment. That challenge, however, was simply a reflection of my lack of experience with using the Protégé software and not a reflection on the module itself.

V. Analysis

The Analysis step of the Gibbs' Reflective Cycle examines lessons learned and identifies opportunities for future improvement (Praveena et al, 2025).

One of the most valuable lessons from the module for my development work was simply demystifying the baseline concept of an ontological model in relation to real world use cases. While I have worked extensively with database models and schemas, for practical business reasons, that work in my career has rarely brought with it the opportunity to consider the formal conceptual foundations for how and why such database-related material is structured in its own right.

A second valuable lesson was developing an awareness of multiple methodologies for developing ontological models, combined with an understanding that no single development approach will be suitable for every use case or every possible subject matter domain. Rather, for maximum productivity, ontological models need to be intentionally and carefully scoped and tailored not only to their subject matter domain or domains, but also to their intended use cases (Iglesias-Molina et al, 2022; Zhang and Xiao, 2024).

Due care should also be given to the practical cost of implementing any given ontological model in code as well, as more robust platforms, such as OWL 2, carry with them the risk of greater resource consumption and software design complexity (Iglesias-Molina et al, 2022; Zhang and Xiao, 2024). In other words, a successful ontological design should combine both accuracy and utility not only in the abstract, conceptual sense, but in the practical engineering sense as well.

VI. Conclusion

The Conclusion step summarizes the conclusions from the preceding steps of the cycle (Praveena et al, 2025). Overall, my conclusion is that a thorough understanding of ontological principles is quite useful for software solutions development, both for AI-driven software, and in general as a fundamental knowledge area underpinning any kind of database-driven software development. By understanding how ontological models are developed in the abstract, translating such abstract conceptual models into working software implementations becomes far more straightforward, accurate, and productive to real world applications.

VII. Action Plan

The Action Plan step of the Gibbs' Reflective Cycle identifies opportunities to enhance learning outcomes and to minimize learning challenges for future study (Praveena et al, 2025).

For future study, I would allocate more time to developing proficiency with the mechanics of using the Protégé platform, as well as researching which platform plugins would be productive to incorporate into ontology model development; for example, plugins for ontology class hierarchy visualizations and for class structure and ontological axiom validation.

I would also spend more time researching, then comparing and contrasting, competing ontological development approaches, with the goal of developing a finer-grained understanding of when any given methodological approach is most appropriate (or contraindicated) for any given subject matter domain and any given software development project, as not every ontological development platform will be practically suitable for every given real world use case (Iglesias-Molina et al, 2022; Zhang and Xiao, 2024).

VIII. References

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